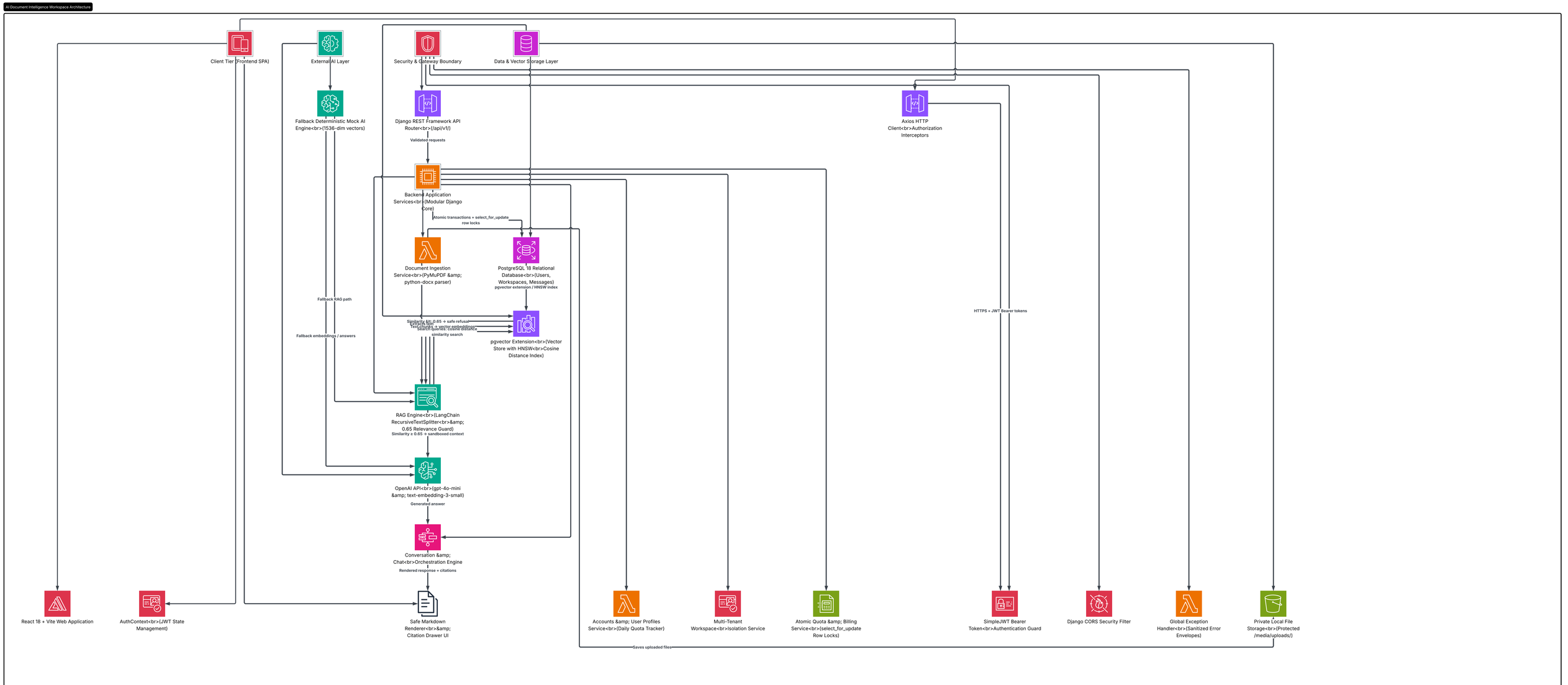

AI Document Intelligence Workspace Architecture
Client Tier (Frontend SPA)
External AI Layer
Security & Gateway Boundary
Data & Vector Storage Layer
Fallback Deterministic Mock AI Engine
(1536-dim vectors)
Django REST Framework API Router
(/api/v1/)
Axios HTTP Client
Authorization Interceptors
Validated requests
Backend Application Services
(Modular Django Core)
Atomic transactions + select_for_update row locks
Document Ingestion Service
(PyMuPDF & python-docx parser)
PostgreSQL 18 Relational Database
(Users, Workspaces, Messages)
pgvector extension / HNSW index
Fallback RAG path
Fallback embeddings / answers
Similarity < 0.65 → safe refusal
Extract text → chunks → vector embeddings
Search pgvector: cosine distance similarity search
pgvector Extension
(Vector Store with HNSW
Cosine Distance Index)
RAG Engine
(LangChain RecursiveTextSplitter
& 0.65 Relevance Guard)
Similarity ≥ 0.65 → sandboxed context
OpenAI API
(gpt-4o-mini & text-embedding-3-small)
Generated answer
Conversation & Chat
Orchestration Engine
Rendered response + citations
HTTPS + JWT Bearer tokens
React 18 + Vite Web Application
AuthContext
(JWT State Management)
Safe Markdown Renderer
& Citation Drawer UI
Accounts & User Profiles Service
(Daily Quota Tracker)
Multi-Tenant Workspace
Isolation Service
Atomic Quota & Billing Service
(select_for_update Row Locks)
SimpleJWT Bearer Token
Authentication Guard
Django CORS Security Filter
Global Exception Handler
(Sanitized Error Envelopes)
Private Local File Storage
(Protected /media/uploads/)
Saves uploaded files